

Numerical Methods for the Wave Equation

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1 The wave equation

The standard one-dimensional linear wave equation has the form

$$\frac{\partial^2 v}{\partial t^2} - c^2 \frac{\partial^2 v}{\partial x^2} = 0. \quad (1.1)$$

One can think of v , for example, as representing the displacement of an infinitesimal piece of a vibrating string from the equilibrium position. This mechanical interpretation makes it evident that suitable initial conditions will concern the initial position and the initial velocity of each infinitesimal piece of the string so that we must associate to (1.1) initial conditions of the form

$$v(x, 0) = f(x), \quad \left. \frac{\partial v}{\partial t} \right|_{t=0} = g(x), \quad (1.2)$$

with f and g given functions. There will also be problem-dependent boundary conditions, e.g.

$$v(x = 0, t) = F_0(t), \quad v(x = L, t) = F_L(t), \quad (1.3)$$

or similar conditions of the Neumann type, i.e., concerning the x -derivatives of v at the-end points rather than the function v itself.

We note that

$$\begin{aligned} \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right) \left(\frac{\partial v}{\partial t} - c \frac{\partial v}{\partial x} \right) &= \frac{\partial}{\partial t} \frac{\partial v}{\partial t} + \frac{\partial}{\partial t} \left(-c \frac{\partial v}{\partial x} \right) + c \frac{\partial}{\partial x} \frac{\partial v}{\partial t} + c \frac{\partial}{\partial x} \left(-c \frac{\partial v}{\partial x} \right) \\ &= \frac{\partial^2 v}{\partial t^2} - c^2 \frac{\partial^2 v}{\partial x^2}. \end{aligned} \quad (1.4)$$

Thus, upon setting

$$\frac{\partial v}{\partial t} - c \frac{\partial v}{\partial x} = u, \quad (1.5)$$

we may write

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0. \quad (1.6)$$

The initial condition for v is given by the first one of (1.2). We obtain the initial condition for u from (1.5) evaluated at $t = 0$:

$$u(x, 0) = \left. \frac{\partial v}{\partial t} \right|_{t=0} - c \frac{\partial}{\partial x} v(x, t=0) = g(x) - c \frac{df}{dx}. \quad (1.7)$$

In this way the original second-order wave equation has been written as an equivalent pair of first-order equations, (1.5) and (1.6). Since these are simpler, we will start by considering the numerical solution of these first-order equations. Before turning to numerics, however, it is useful to consider some analytical aspects of the problem.

2 Some analytical solutions of the wave equations

The wave equation has a simple, exact and elegant general solution found by D'Alembert; it is

$$v(x, t) = F(x - ct) + G(x + ct), \quad (1.1)$$

with the functions F and G depending on the initial conditions, as we show later. To prove that (1.1) is indeed a solution we note that $F(x - ct)$ is really a function of a *single* variable, the “package” $z = x - ct$, $F = F(z(x, t))$. Thus, using the differentiation rule for a function of a function, we have

$$\frac{\partial F}{\partial x} = \frac{\partial z}{\partial x} \frac{dF}{dz} = \frac{dF}{dz}, \quad \frac{\partial F}{\partial t} = \frac{\partial z}{\partial t} \frac{dF}{dz} = -c \frac{dF}{dz}, \quad (1.2)$$

in which dF/dz is the derivative of F with respect to the single argument on which it depends, i.e., the entire “package” $x - ct$. For example,

$$\frac{\partial}{\partial x} \sin(x - ct) = \cos(x - ct) \quad \frac{\partial}{\partial t} \sin(x - ct) = -c \cos(x - ct). \quad (1.3)$$

In the same way we find

$$\frac{\partial^2 F}{\partial x^2} = \frac{\partial z}{\partial x} \frac{d}{dz} \left(\frac{dF}{dz} \right) = \frac{d^2 F}{dz^2}, \quad \frac{\partial^2 F}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial F}{\partial t} \right) = \frac{\partial}{\partial t} \left[-c \frac{dF}{dz} \right] = \frac{\partial z}{\partial t} \frac{d}{dz} \left[-c \frac{dF}{dz} \right] = c^2 \frac{d^2 F}{dz^2}, \quad (1.4)$$

so that, indeed, F satisfies (1.2). A similar calculation proves that G is also a solution. The proof that (1.1) is the most general solution is more difficult and we will not attempt to present it.

Let us now show how F and G are related to the initial conditions (1.2). For simplicity, let us take $g = 0$. Then we have

$$v(x, t = 0) = f(x) = F(x) + G(x), \quad \left. \frac{\partial v}{\partial t} \right|_{t=0} = -cF'(x) + cG'(x) = 0. \quad (1.5)$$

From the second equation we have $G(x) = F(x) + K$ for some constant K and, substituting into the first one,

$$F(x) + F(x) + K = f(x), \quad (1.6)$$

from which

$$F = \frac{1}{2}(f - K), \quad G = \frac{1}{2}(f - K) + K = \frac{1}{2}(f + K). \quad (1.7)$$

Upon adding F and G to form D'Alembert solution (1.1) the constant K drops out and we find

$$v(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)]. \quad (1.8)$$

A similar argument can be applied to the more general case in which $g \neq 0$.

The function $F(x - ct)$ describes a perturbation propagating in the direction of increasing x with speed c (assumed positive) maintaining exactly the same shape for all times. To show this we simply change the original reference frame to one moving with a velocity c . This change is effected by writing $x = x' + ct$, where x' is the spatial coordinate in the moving frame. Evidently $F(x - ct) = F(x')$, which says that, observed from the moving frame, the shape of the wave is independent of time so that it remains identical to the initial shape. In other words, the particular value of F which was at the point x_1 at the initial time $t = 0$ has moved, at the time t , to the position $x'_1 = x_1 + ct$: the “trajectory” of the initial value $F(x_1)$ is the line $x_1 + ct$. In the same way we can show that G represents a perturbation propagating in the direction of decreasing x (for $c > 0$) maintaining exactly the same shape for all times. Thus, the general solution of the wave equation consists of the sum of two perturbations propagating in opposite directions, as if the initial conditions (1.2) were to “split” into two counter-propagating disturbances.

The first image that comes to mind thinking of a vibrating string is not that of propagating waves, but an oscillatory motion in which each point goes up and down oscillating with the same frequency but different amplitudes. To show how this situation is actually contained in (1.1) let us consider the special case

$$F(x - ct) = A_0 \cos[k(x - ct)], \quad G(x + ct) = A_0 \cos[k(x + ct)]. \quad (1.9)$$

Then

$$\begin{aligned} v(x, t) &= A_0 (\cos[k(x - ct)] + \cos[k(x + ct)]) = 2A_0 \cos\left(k \frac{x - ct + x + ct}{2}\right) \cos\left(k \frac{x - ct - x - ct}{2}\right) \\ &= 2A_0 \cos(kx) \cos(kct). \end{aligned} \quad (1.10)$$

What this relation shows is that a point x_1 oscillates with an angular frequency $\omega = ct$ and an amplitude $2A_0 \cos(kx_1)$, while another point x_2 oscillates with the same angular frequency and an amplitude $2A_0 \cos(kx_2)$. A solution of this type is called a *standing wave* and, on the surface, the very notion of propagating waves seems alien to it. However, from its derivation, we clearly see that the standing wave is really the result of the superposition of counter-propagating waves.

3 Discretization of the first-order wave equation

A priori we can imagine many different ways in which (1.6) can be discretized; limiting ourselves to first-order accuracy in time we may write, for example:

- Backward Euler, first-order in x , explicit in time

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_j^n - u_{j-1}^n}{\Delta x} = 0, \quad (3.1)$$

- Backward Euler, first-order in x , implicit in time

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_j^{n+1} - u_{j-1}^{n+1}}{\Delta x} = 0, \quad (3.2)$$

- Forward Euler, first-order in x , explicit in time

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_{j+1}^n - u_j^n}{\Delta x} = 0 \quad (3.3)$$

- Forward Euler, first-order in x , implicit in time

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_{j+1}^{n+1} - u_j^{n+1}}{\Delta x} = 0 \quad (3.4)$$

- Centered difference, second-order in x , explicit in time

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0, \quad (3.5)$$

- Centered difference, second-order in x , implicit in time

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} = 0, \quad (3.6)$$

and others as well. Here, as usual, we have set $u_j^n = u(x_j, t^n)$. If we wanted second-order accuracy in both space and time we might want to try

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + c \frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} = 0, \quad (3.7)$$

which is implicit, or the explicit analog of this. Our task is now to see which of these methods is preferable in terms of accuracy and stability.

4 The method of effective equations

There is a very nice way to gain a deep insight into the properties of the various possible discretizations, which is called the method of effective equations. The idea is that, when we use any one of the previous discretizations, what we are solving is not really (1.6) but an equation that differs from it because of the presence of errors that are a consequence of the discretization. For example, by using the usual Taylor series argument, we see that

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{1}{2}\Delta x \frac{\partial^2 u}{\partial x^2} + \dots \quad (4.1)$$

and

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\partial u}{\partial t} + \frac{1}{2}\Delta t \frac{\partial^2 u}{\partial t^2} + \dots \quad (4.2)$$

This latter result can be re-expressed noting that, from (1.6),

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial t} \right) = \frac{\partial}{\partial t} \left(-c \frac{\partial u}{\partial x} \right) = -c \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right) = c^2 \frac{\partial^2 u}{\partial x^2}. \quad (4.3)$$

Upon substituting (4.1) and (4.3) into (3.1) and rearranging we see that (3.1) is not really the same as the original equation (1.6), but (up to higher order terms) is closer to

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = \frac{c\Delta x}{2} \left(1 - \frac{c\Delta t}{\Delta x} \right) \frac{\partial^2 u}{\partial x^2}. \quad (4.4)$$

Sure enough, when Δx and Δt tend to zero, the error vanishes, which proves that the discretization (3.1) is consistent with (1.6) but, for any finite Δx and Δt , what we are really solving is (4.4) rather than (1.6). Alternatively put, we may say that what we find by numerically solving (3.1) will be closer to the solution of (4.4) than to the solution of the original equation (1.6). The use of the effective equation (4.4) to study the properties of the various possible discretizations of (1.6) gives this method its name.

If we set

$$D = \frac{c\Delta x}{2} \left(1 - \frac{c\Delta t}{\Delta x} \right), \quad (4.5)$$

(4.4) becomes

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2}. \quad (4.6)$$

we see that (4.4) contains a diffusive process of purely numerical origin (in the right-hand side), in addition to the original propagation process (in the left-hand side).¹

In order to see what are the consequences of the non-zero right-hand side of (4.6), we again consider a special solution (i.e., a single Fourier mode) corresponding to the initial condition

$$f(x) = e^{ikx}, \quad (4.7)$$

with i the imaginary unit and k a constant.² It is readily verified that the exact solution of (1.6) subject to this initial condition is simply

$$u_{ex}(x, t) = e^{ik(x-ct)}. \quad (4.8)$$

¹If we effect the change of variables $x = x + ct$ mentioned before in section 2, (4.6) becomes the diffusion equation in the form seen before, namely

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x'^2}.$$

As seen from the moving frame, therefore, the wave shape will not remain constant but diffuse away like, for example, a spot of high temperature in a heat-conducting material.

² k is the wave number, related to the wave length λ by $k = 2\pi/\lambda$; the relation is similar to that between the angular frequency ω and the period T , which is given by $\omega = 2\pi/T$.

To find the solution of the effective equation (4.6) subject to the same initial condition we try

$$u(x, t) = G(t)e^{ikx}, \quad \text{with} \quad G(0) = 1. \quad (4.9)$$

Upon substitution into (4.6) we have

$$\frac{dG}{dt} + ikcG = D(-k^2)G, \quad (4.10)$$

which is readily integrated by separating the variables

$$\frac{dG}{G} = -(ikc + Dk^2)dt, \quad (4.11)$$

to find, after imposing the initial condition $G(0) = 1$,

$$G(t) = e^{-k(ic + kD)t}. \quad (4.12)$$

Upon substituting into (4.9) we find then that the solution of (4.6) is given by

$$u(x, t) = e^{-Dk^2t}e^{ik(x-ct)}. \quad (4.13)$$

The second factor is the exact solution, but it is seen that its amplitude, instead of remaining 1 for all time as in (4.8), depends on time according to the first factor. If $D > 0$, the amplitude decreases, the slower the smaller Δx (cf. (4.5), but it decreases nonetheless. This is a typical instance of *numerical diffusion* or *artificial dissipation*. The rate of decay is greater the larger k , i.e., short wavelengths get damped faster than long ones. The situation is much worse if $D < 0$ because, in this case, the amplitude grows without bound: the numerical method that we have used is unconditionally unstable and produces a physically unacceptable result. For stability we must therefore require that $D \geq 0$. If $c > 0$ (waves moving toward increasing x) the stability condition is therefore

$$C \equiv \frac{c\Delta t}{\Delta x} \leq 1. \quad (4.14)$$

The quantity C in the left-hand side is known as the *Courant number* and this relation is known as the *CFL condition* (short for Courant, Friedrichs, Levy). The definition (4.5) of D shows that $D = 0$ when the Courant number is exactly equal to 1. This feature is related to the special nature of the lines $x \pm ct$ which are the *characteristic lines* of the equation. However exploiting this fact is only possible in the very simple case considered here in which c is a constant. In a real-life application the quantity analogous to c would be different from place to place and time to time and, therefore, it is usually impossible to choose a discretization that guarantees a unit Courant number in the entire domain of the calculation for all times.

We also see from (4.5) that $D < 0$ for $c < 0$ no matter how Δt and Δx are chosen, so that the discretization (3.1) is unconditionally unstable for left-propagating waves.

The same analysis can be carried out for all the other discretizations shown in section 3. In particular, if we apply it to the explicit forward Euler discretization (3.3), we find the same equation (4.6) in which D is given by an expression that we write in such a way as to make comparison with (4.5) easier:

$$D = \frac{(-c)\Delta t}{\Delta x} \left(1 - \frac{(-c)\Delta t}{\Delta x} \right). \quad (4.15)$$

If $c > 0$ D will be negative irrespective of the choice of Δt and Δx and therefore the method will be unstable. However, if $c < 0$, the calculation is stable provided that

$$\frac{|c|\Delta t}{\Delta x} \leq 1. \quad (4.16)$$

We thus have found something rather peculiar: for waves propagating toward increasing x it is necessary to use the backward Euler method to obtain stable solutions, while the forward Euler method is necessary for waves propagating in the opposite direction; in either cases the Courant number cannot exceed 1. This conclusion is at the root of what is called *upwind differencing*: the spatial derivative must be discretized using the node “upwind” (i.e., in the direction opposite to the “wind velocity” c) of the node of interest, namely u_{j-1} if $c > 0$ and u_{j+1} if $c < 0$.

5 Dispersive error

Let us keep one more term in the expansions (4.1) and (4.3) for $\partial u/\partial x$ and $\partial u/\partial t$:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\partial u}{\partial t} + \frac{1}{2}\Delta t \frac{\partial^2 u}{\partial t^2} + \frac{1}{6}\Delta t^2 \frac{\partial^3 u}{\partial t^3} + \dots, \quad (4.17)$$

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{1}{2}\Delta x \frac{\partial^2 u}{\partial x^2} + \frac{1}{6}\Delta x^2 \frac{\partial^3 u}{\partial x^3} + \dots. \quad (4.18)$$

By proceeding as before we can re-express the time derivatives in the first equation in terms of space derivatives finding

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\partial u}{\partial t} + \frac{1}{2}\Delta t c^2 \frac{\partial^2 u}{\partial x^2} - \frac{1}{6}\Delta t^2 c^3 \frac{\partial^3 u}{\partial x^3} + \dots. \quad (4.19)$$

Now the modified equation is

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2} + cE \frac{\partial^3 u}{\partial x^3}, \quad (4.20)$$

with D given by (4.5) and

$$E = \frac{\Delta x^2}{6} \left(1 - \frac{\Delta t^2 c^2}{\Delta x^2} \right). \quad (4.21)$$

Proceeding in the usual way we find

$$\frac{dG}{dt} = -(ikc + k^2 D + ik^3 cE)G, \quad (4.22)$$

from which, with $G(0) = 1$, we find

$$G(t) = \exp[-ikc(1 + Ek^2)t] e^{-k^2 Dt}, \quad (4.23)$$

from which

$$u = \exp[ikx - ic(1 + Ek^2)t] e^{-k^2 Dt}. \quad (4.24)$$

Upon comparison with the exact solution (4.8) we notice two significant differences. The first one is the damping of the wave proportionally to $\exp(-k^2 Dt)$ as found before. The new feature is the fact that the speed of propagation is not the constant c independent of k , but now has a k -dependence and has the effective value

$$c_{eff}(k) = (1 + Ek^2)c. \quad (4.25)$$

Since $E > 0$ in order to have $D > 0$, we see that waves with large k (small wavelengths) race ahead of those with smaller k 's if. If we consider a wave consisting at $t = 0$ of the superposition of several k 's:

$$u(x, 0) = \sum_n c_n e^{ik_n x}, \quad (4.26)$$

at time $t > 0$ it will be given by

$$u(x, t) = \sum_n c_n e^{i[k_n x - c(1 + Ek_n^2)t]} e^{-k_n^2 Dt}. \quad (4.27)$$

In addition to the damping effect, we see that the various components propagate at different velocities and therefore acquire a phase difference with respect to one other. The end result is that the wave changes shape. This form of error is called *dispersive* because it leads to a “dispersion” of the wave shape.

6 Higher-order schemes

The diffusive behavior that we have encountered is due to the first-order error of the discretizations we have used for the derivatives. As far as the space derivative is concerned, we can improve the accuracy by using the centered formula. A way to improve the time discretization is the following.

From the Taylor series expansion

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\partial u}{\partial t} + \frac{1}{2}\Delta t \frac{\partial^2 u}{\partial t^2} + O(\Delta t^2) \quad (4.28)$$

we see that, if we were to approximate $\partial u / \partial t$ as

$$\frac{\partial u}{\partial t} = \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{1}{2}\Delta t \frac{\partial^2 u}{\partial t^2} + \dots \quad (4.29)$$

the error would become of order Δt^2 rather than Δt . As we have already remarked after (4.3) this can also be written

$$\frac{\partial u}{\partial t} = \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{1}{2}\Delta t c^2 \frac{\partial^2 u}{\partial x^2} + \dots \quad (4.30)$$

The idea now is to use the usual centered difference formula to express the second space derivative writing

$$\frac{\partial u}{\partial t} \simeq \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{\Delta t c^2}{2} \frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{\Delta x^2}. \quad (4.31)$$

Upon substituting this relation for $\partial u / \partial t$ and using the centered formula for $\partial u / \partial x$ in the equation (1.6) we then find

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{\Delta t c^2}{2} \frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{\Delta x^2} + c \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0. \quad (4.32)$$

Upon reorganizing this relation we have the explicit *Lax-Wendroff* scheme:

$$u_j^{n+1} = u_j^n - \frac{c\Delta t}{2\Delta x} (u_{j+1}^n - u_{j-1}^n) + \frac{c^2\Delta t^2}{2\Delta x^2} (u_{j-1}^n - 2u_j^n + u_{j+1}^n). \quad (4.33)$$

The usual procedure applied to this equation gives the modified equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = \frac{c\Delta x^2}{6} \left(1 - \frac{c^2\Delta t^2}{\Delta x^2} \right) \frac{\partial^3 u}{\partial x^3}. \quad (4.34)$$

Remarkably, the damping effect is absent (of course, it is present at higher order, and therefore it is less troublesome) and the dominant error is dispersive. The stability condition is found to be the usual one given in (4.14).

7 The second-order wave equation

We have seen that the second-order wave equation (1.1) can be split into the system (1.5) and (1.6) and this system can form the basis for a numerical method. On the assumption that $c > 0$ (which is no restriction since only c^2 appears in the equation), a simple discretization would be

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_j^n - u_{j-1}^n}{\Delta x} = 0, \quad \frac{v_j^{n+1} - v_j^n}{\Delta t} - c \frac{v_{j+1}^n - v_j^n}{\Delta x} = u_j^n. \quad (4.35)$$

Notice that we have used upwind differencing in both space derivatives. Stability of this method requires that $c\Delta t < \Delta x$.

The method can be improved by using the Lax-Wendroff idea based on (4.30):

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\partial u}{\partial t} - \frac{1}{2}\Delta t \frac{\partial^2 u}{\partial t^2} + \dots, \quad \frac{v_j^{n+1} - v_j^n}{\Delta t} = \frac{\partial v}{\partial t} - \frac{1}{2}\Delta t \frac{\partial^2 v}{\partial t^2} + \dots \quad (4.36)$$

As before we see from the first one of (4.35) that $\partial_t^2 u = c^2 \partial_x^2 u$ and we see directly from the wave equation that the same relation is satisfied by v , namely $\partial_t^2 v = c^2 \partial_x^2 v$. Thus, as before, we have the scheme

$$u_j^{n+1} = u_j^n - \frac{c\Delta t}{2\Delta x}(u_{j+1}^n - u_{j-1}^n) + \frac{c^2\Delta t^2}{2\Delta x^2}(u_{j-1}^n - 2u_j^n + u_{j+1}^n). \quad (4.37)$$

$$v_j^{n+1} = v_j^n + \frac{c\Delta t}{2\Delta x}(v_{j+1}^n - v_{j-1}^n) + \frac{c^2\Delta t^2}{2\Delta x^2}(v_{j-1}^n - 2v_j^n + v_{j+1}^n) + u_j^n. \quad (4.38)$$

The second-order equation can also be discretized directly. The simplest discretization is

$$\frac{v_j^{n+1} - 2v_j^n + v_j^{n-1}}{\Delta t^2} = c^2 \frac{v_{j-1}^n - 2v_j^n + v_{j+1}^n}{\Delta x^2}. \quad (4.39)$$

This is an explicit method, in that it gives directly v_j^{n+1} if the solution at the two previous time levels is known. Not surprisingly, it is found that this scheme is stable if $c\Delta t \leq \Delta x$.

The method requires a special treatment of the first time step $t^1 = \Delta t$ as can be seen by setting $n = 0$ in (4.39):

$$\frac{v_j^1 - 2v_j^0 + v_j^{-1}}{\Delta t^2} = c^2 \frac{v_{j-1}^0 - 2v_j^0 + v_{j+1}^0}{\Delta x^2}. \quad (4.40)$$

Here $v_j^0 = f(x_j)$ is just the initial condition from (1.2), but v_j^{-1} , i.e., v_j at time $t = -\Delta t$, is undefined. We can handle this problem by adapting to the case of the time variable the ghost node idea we that we used for the space variable in the case of Neumann type boundary conditions. We write

$$\frac{v_j^1 - v_j^{-1}}{2\Delta t} = g_j, \quad (4.41)$$

where the function $g_j = g(x_j)$ is known from the initial conditions (1.2). From here we have

$$v_j^{-1} = v_j^1 - 2\Delta t g_j. \quad (4.42)$$

In this way (4.40) becomes

$$\frac{v_j^1 - 2f_j + (v_j^1 - 2\Delta t g_j)}{\Delta t^2} = c^2 \frac{f_{j-1}^0 - 2f_j^0 + f_{j+1}^0}{\Delta x^2}, \quad (4.43)$$

or

$$\frac{2v_j^1 - 2f_j - 2\Delta t g_j}{\Delta t^2} = c^2 \frac{f_{j-1}^0 - 2f_j^0 + f_{j+1}^0}{\Delta x^2}. \quad (4.44)$$

The next time step with $n = 1$ will involve v_j^1 , given by this relation, and $v_j^0 = f_j$. The relation (4.40) applies to all the following time steps without modification.

A more stable method can be based on an idea similar to the Crank-Nicolson discretization of the diffusion equation, namely

$$\frac{v_j^{n+1} - 2v_j^n + v_j^{n-1}}{\Delta t^2} = \frac{c^2}{2} \left(\frac{v_{j-1}^{n+1} - 2v_j^{n+1} + v_{j+1}^{n+1}}{\Delta x^2} + \frac{v_{j-1}^{n-1} - 2v_j^{n-1} + v_{j+1}^{n-1}}{\Delta x^2} \right). \quad (4.45)$$

It is found that the modulus of the amplification factor for this scheme equals 1 whatever the choice of Δt and Δx , so that the method is never unstable. As before, when this equation is written for $n = 1$, the last three terms in the right-hand side introduce values of v at the unphysical time $-\Delta t$. This issue can be handled by means of ghost (time) nodes as before.